

Say, Echo, Do: Strategic Speech, Media Echo, and the Economics of False Alarms

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Abstract

Markets hear three voices: what institutions *say*, what the media *echoes*, and what capital *does*. We develop a theory of their disagreement, centred on an endogenous account of *false alarms*. An institution that trades while addressing a partly credulous crowd optimally talks an asset down while buying it exactly when $\varphi^2 < 2\lambda k < \varphi$ (credulity φ , price impact λ , misreporting cost k), and exaggerates while selling when $\varphi > 1$. A distribution-free identity, $\text{Cov}(r, \text{Say}) = \lambda \text{Cov}(\text{Do}, \text{Say}) - \varphi \sigma_\varepsilon^2$, makes the observable covariance between words and deeds the switch that decides whether words are followed or faded, and adaptive trust undergoes a flip bifurcation at $\lambda k = 1$ that yields closed-form credibility cycles and, through period-doubling, chaos. For measurement, we prove an exact *semantic declustering* theorem that separates news from echo using timing and embedding similarity, show that return-aligned contrastive embeddings are affine isometries of market-outcome space with a Pinsker certificate for imperfect training, and show that Lévy area identifies whether money moved before the narrative, with every closed form confirmed by Monte Carlo simulation.

1. Introduction

A respected bank publishes a gloomy call. Headlines multiply. The stock falls, though by less than the noise suggests, and weeks later it trades higher. In hindsight, someone was buying all along (Figure 1a). Practitioners call such episodes *false alarms*. Proving intent is neither possible nor necessary. What is possible is to measure the disagreement between three observable voices: **Say** (institutional statements such as ratings, notes and management tone), **Echo**

(media repetition and amplification) and **Do** (revealed allocation: holdings, insider trades, options flow and price).

We develop a theory of that disagreement and of how to trade it. Two principles organise the paper. *Deeds over words*: when **Say** and **Do** disagree, the optimal forecast is proportional to their weighted gap, and the *sign* of their covariance decides whether words are followed or faded. *Fade the echo, follow the news*: repetition carries no information, so its price effect must unwind, whereas under-weighted news drifts. The theory never prescribes blind contrarianism; it identifies which cell of Figure 1b applies.

Contributions. The results we believe to be new are:

- **Endogenous false alarms** (Section 4). A closed-form theory of optimal speech while trading, with a four-regime taxonomy: truth, shading, false alarm, and exaggeration with distribution (Theorem 4.3 and Corollary 4.4).
- **The Say–Do covariance identity** (Theorem 4.5), which holds for any distribution and turns an observable statistic into a follow-or-fade switch.
- **A manipulation dichotomy under virality** (Proposition 4.6) and **credibility cycles** under adaptive trust (Theorem 4.7), with explicit bifurcation thresholds $\lambda k = 1$ and $\lambda k = \frac{5}{6}$.
- **Semantic declustering** (Theorem 5.3): exact, factorised news-versus-echo posteriors that combine timing with embedding similarity.
- **Embedding geometry with a certificate** (Theorems 5.4 and 5.5), **absorption as an interpretation gap** (Theorem 3.7), and **lead–lag via Lévy area** (Theorem 5.6).

The Gaussian follow-or-fade and Say–Do gap results (Theorems 3.3 and 3.6) are clean multi-channel statements of Bayesian updating. Regime detection, conformal calibration and Kelly sizing (Appendix B) are classical tools, included so that the paper is self-contained. Proofs are in Appendix A.

2. Related Work

Sentiment and returns. Media pessimism predicts price pressure followed by reversal (Tetlock, 2007). Financial

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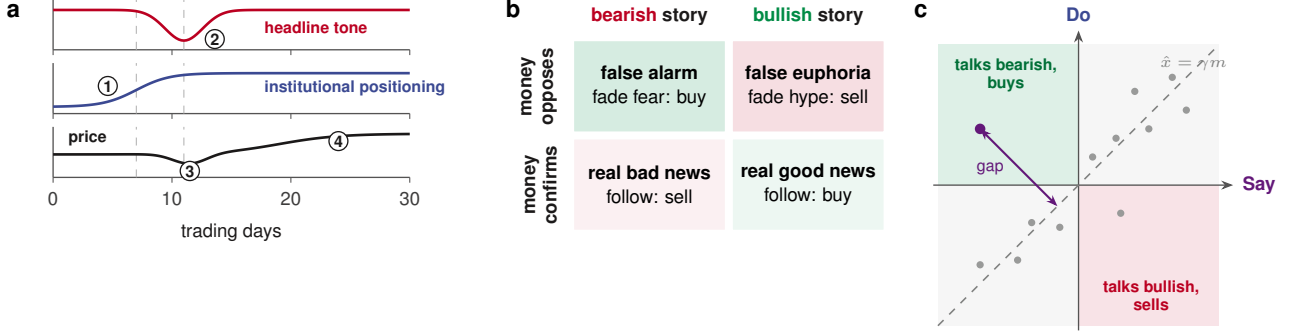


Figure 1. **Three voices, one decision.** **a**, The false-alarm pattern (stylised): positioning builds (1) before the alarm (2); price falls far less than the alarm implies (3), then recovers (4). **b**, The narrative never decides the trade on its own; its agreement with money and price does. **c**, The Say-Do plane. The optimal forecast decreases with signed distance from the line $\hat{x} = \gamma m$ (Theorem 3.6); the top-left quadrant is the false-alarm zone.

text needs domain-specific measurement (Loughran & McDonald, 2011), sentiment matters most where arbitrage is hard (Baker & Wurgler, 2006; Shleifer & Vishny, 1997; De Long et al., 1990), and attention drives demand (Barber & Odean, 2008; Da et al., 2011). Prices drift after headlines and reverse after no-news moves (Chan, 2003), heterogeneous agents reconcile under- and overreaction (Hong & Stein, 1999), and the same news is read differently across states (Veronesi, 1999). Text factors and language models predict returns (Ke et al., 2019; Araci, 2019; Lopez-Lira & Tang, 2023), subject to look-ahead bias (Glasserman & Lin, 2023).

Informed trading and communication. Order flow reveals private information (Kyle, 1985; Glosten & Milgrom, 1985). Volume–return dynamics separate informed from hedging trades (Llorente et al., 2002), predatory traders exploit predictable flow (Brunnermeier & Pedersen, 2005), short sellers process news better (Engelberg et al., 2012), insider trades split into informative and routine (Cohen et al., 2012), and promotional articles move prices (Kogan et al., 2023). Cheap talk (Crawford & Sobel, 1982), costly talk with credulous receivers (Kartik et al., 2007), manipulation by informed “gurus” (Bénabou & Laroque, 1992) and rumour-based trading (van Bommel, 2003) study how informed agents shape beliefs.

Mathematical tools. We use Hawkes processes (Hawkes, 1971) and their cluster representation (Hawkes & Oakes, 1974), which underlies stochastic declustering (Zhuang et al., 2002), with finance applications in (Bacry et al., 2015). We also use contrastive learning (van den Oord et al., 2018; Reimers & Gurevych, 2019) and the linear representation hypothesis (Park et al., 2024); Lévy area and path signatures (Lévy, 1940; Lyons, 1998; Chevyrev & Kormilitzin, 2016); cobweb learning and period-doubling (Ezekiel, 1938; Evans & Honkapohja, 2001; May, 1976); and the square-root impact law (Bouchaud et al., 2018).

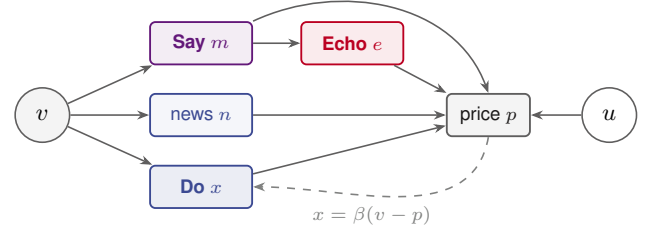


Figure 2. **Information structure.** Echo is a downstream garbling of institutional speech, news is an independent signal of value, and informed demand responds to the price (dashed).

3. A Three-Voice Market

Assumption 3.1 (Signals). Value is $v \sim \mathcal{N}(0, \sigma_v^2)$, and public channels are $\mathbf{y} = \mathbf{h}v + \boldsymbol{\nu} \in \mathbb{R}^K$ with $\boldsymbol{\nu} \sim \mathcal{N}(\mathbf{0}, \Sigma_\nu)$ independent of v and $\Sigma_y = \sigma_v^2 \mathbf{h}\mathbf{h}^\top + \Sigma_\nu \succ 0$.

Assumption 3.2 (Prices). Noise demand $u \sim \mathcal{N}(0, \sigma_u^2)$ is independent of everything else. Informed demand is $x = \beta(v - p)$ with $\beta > 0$, and the price is $p = \boldsymbol{\varphi}^\top \mathbf{y} + \lambda(x + u)$ with $\lambda > 0$, where $\boldsymbol{\varphi}$ holds the crowd’s narrative multipliers.

Write $\kappa = (1 + \lambda\beta)^{-1}$, $\mathbf{w}^* = \sigma_v^2 \Sigma_y^{-1} \mathbf{h}$ (the Bayes weights, $\mathbb{E}[v | \mathbf{y}] = \mathbf{w}^{*\top} \mathbf{y}$), $\sigma_{v|y}^2 = \text{Var}(v | \mathbf{y})$, and $r = v - p$. Solving the fixed point gives $r = \kappa(v - \boldsymbol{\varphi}^\top \mathbf{y} - \lambda u)$ (Figure 2).

3.1. Follow or fade

Theorem 3.3 (Follow-or-fade). Under Assumptions 3.1 and 3.2, $\mathbb{E}[r | \mathbf{y}] = \kappa(\mathbf{w}^* - \boldsymbol{\varphi})^\top \mathbf{y}$ and $\text{Var}(r | \mathbf{y}) = \kappa^2(\sigma_{v|y}^2 + \lambda^2 \sigma_u^2) =: \varsigma^2$. The loading on channel k is negative (fade) iff $\varphi_k > w_k^*$.

Corollary 3.4 (Fade the echo). If $e = \mathbf{a}^\top \mathbf{y}_{-e} + \xi$ with ξ independent noise of positive variance, then $w_e^* = 0$, and the loading on e is exactly $-\kappa\varphi_e$.

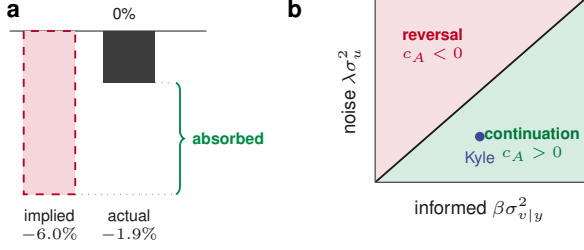


Figure 3. **Absorption.** **a**, The narrative implied a 6% fall; the price fell 1.9%. **b**, Phase diagram of Theorem 3.7: absorption signals a hidden buyer only where informed intensity dominates noise.

Corollary 3.5 (Follow the news). *If $n = v + \nu_n$ with independent noise ν_n , then $w_n^* \in (0, 1)$, and returns drift with n whenever $\varphi_n < w_n^*$.*

3.2. Deeds over words

Theorem 3.6 (Say–Do gap optimality). *Suppose a positioning proxy $\hat{x} = x + \xi_x$ with $\text{Var } \xi_x = \sigma_x^2$ is also observed. Let $c_x = \beta\zeta^2/(\beta^2\zeta^2 + \sigma_x^2)$ and $\theta = \sigma_x^2/(\beta^2\zeta^2 + \sigma_x^2)$. Then*

$$\mathbb{E}[r | \mathbf{y}, \hat{x}] = \theta\kappa(w^* - \varphi)^\top \mathbf{y} + c_x \hat{x}, \quad \text{Var}(r | \mathbf{y}, \hat{x}) = \theta\zeta^2.$$

*If the **Say** channel $m = y_1$ is overweighted ($\varphi_1 > w_1^*$), the forecast depends on $(m, \hat{x})$ only through $-c_x(\gamma m - \hat{x})$, with $\gamma = \theta\kappa(\varphi_1 - w_1^*)/c_x > 0$.*

3.3. Absorption: informed or noise?

The *absorption* of a narrative is the price move net of the narrative-implied move, $a = p - \varphi^\top \mathbf{y}$.

Theorem 3.7 (Absorption sign). $\mathbb{E}[r | \mathbf{y}, a] = \mathbb{E}[r | \mathbf{y}] + c_A(a - \mathbb{E}[a | \mathbf{y}])$ with

$$c_A = \frac{\beta\sigma_{v|y}^2 - \lambda\sigma_u^2}{\lambda(\beta^2\sigma_{v|y}^2 + \sigma_u^2)}.$$

Absorption predicts continuation iff $\beta\sigma_{v|y}^2 > \lambda\sigma_u^2$, and reversal otherwise. The limits are $c_A = 1/(\lambda\beta)$ as $\sigma_u \rightarrow 0$ and $c_A = -1$ as $\beta \rightarrow 0$.

Lemma 3.8 (Robustness). *For any $\hat{\varphi}, \hat{a} = p - \hat{\varphi}^\top \mathbf{y}$ generates the same σ -field as a given $\mathbf{y}$, so $\mathbb{E}[r | \mathbf{y}, \hat{a}] = \mathbb{E}[r | \mathbf{y}, a]$.*

At Kyle’s equilibrium intensities (Kyle, 1985), $c_A = \frac{1}{2}$. Theorem 3.7 is in the spirit of Llorente et al. (2002), but it conditions on the narrative (Figure 3). In practice, the narrative-implied move comes from the embedding-based interpretation model of Section 5.2, and Lemma 3.8 protects the signal against misestimating it.

4. Strategic Speech and False Alarms

We now let the institution *choose* what to say.

Assumption 4.1 (Strategic institution). An institution knows v , which may have any distribution with mean 0 and variance $\sigma_v^2 < \infty$. It publishes m and trades x . The public sees $\tilde{m} = m + \varepsilon$, where ε has mean zero and variance σ_ε^2 and is independent of v . The price is $p = \varphi\tilde{m} + \lambda x$, where $\varphi \geq 0$ is the crowd’s *credulity*. Misreporting costs $\frac{k}{2}(m - v)^2$. The institution maximises $J = x(v - \varphi m - \lambda x) - \frac{k}{2}(m - v)^2$. Let $a := 2\lambda k$.

Assumption 4.2 (Echo amplification). Each article moves the price by φ_0 per unit of message, and a Hawkes cascade with branching ratio $n \in [0, 1)$ repeats it. Hence $\varphi = \varphi_0/(1 - n)$ (Proposition 5.1).

Theorem 4.3 (Optimal speech and trade). *J has a unique maximiser iff $a > \varphi^2$, namely $m^* = \psi v$ and $x^* = \chi v$ with*

$$\psi = \frac{a - \varphi}{a - \varphi^2}, \quad \chi = \frac{k(1 - \varphi)}{a - \varphi^2}.$$

If $a < \varphi^2$, J is unbounded: only position limits contain manipulation.

Corollary 4.4 (Taxonomy). *Let $a > \varphi^2$ and $v > 0$. The institution tells the **truth** if $\varphi = 1$. It **shades** ($0 < \psi < 1$, buys) if $\varphi < 1 < a/\varphi$. It raises a **false alarm** ($\psi < 0 < \chi$: talks down, buys) if $\varphi^2 < a < \varphi < 1$. It **exaggerates and sells** ($\psi > 1$, $\chi < 0$) if $\varphi > 1$.*

Theorem 4.5 (Say–Do covariance identity). *Under Assumption 4.1, let the message rule $m(v)$ be arbitrary and the trade optimal given it. Then, pathwise, $r = \lambda x - \varphi\varepsilon$. Hence, for any law of v ,*

$$\text{Cov}(r, \tilde{m}) = \lambda \text{Cov}(x, \tilde{m}) - \varphi\sigma_\varepsilon^2, \quad \mathbb{E}[r | x] = \lambda x.$$

In equilibrium, $\text{Cov}(x, \tilde{m}) = \psi\chi\sigma_v^2$, which is negative exactly in the false-alarm and exaggeration regimes.

Why it matters. Both sides of the identity are estimable, and no distributional assumption is needed. The rolling sign of $\text{Cov}(\text{Say}, \text{Do})$ tells the econometrician whether to follow or fade institutional words, while deeds are always followed, with slope λ .

Proposition 4.6 (Manipulation dichotomy). *Under Assumptions 4.1 and 4.2 with $\varphi_0 \in (0, 1)$, call a regime manipulative if $\text{Cov}(\text{Say}, \text{Do}) < 0$. (i) If $a < 1$ (a deep market), the only manipulative regime is the false alarm, which occurs iff $\max\{0, 1 - \varphi_0/a\} < n < 1 - \varphi_0/\sqrt{a}$. (ii) If $a > 1$ (a shallow market), it is exaggeration, which occurs iff $1 - \varphi_0 < n < 1 - \varphi_0/\sqrt{a}$. (iii) For $n > 1 - \varphi_0/\sqrt{a}$ no interior optimum exists.*

Adaptive trust. Suppose the crowd resets its credulity each round to the least-squares slope of value on message under the previous strategy, a cobweb rule (Ezekiel, 1938; Evans & Honkapohja, 2001). This gives $\varphi_{t+1} = 1/\psi(\varphi_t) = F_a(\varphi_t)$ with $F_a(\varphi) = (a - \varphi^2)/(a - \varphi)$.

Theorem 4.7 (Credibility cycles). (i) For $a > 1$, truth ($\varphi = 1$) is the unique admissible fixed point. (ii) $F'_a(1) = -1/(a-1)$, so truth is stable iff $\lambda k > 1$. (iii) For $1 < a < 2$ there is exactly one 2-cycle,

$$\varphi_{\pm} = \frac{1}{2}(a \pm \sqrt{a(2-a)}), \quad \varphi_- < 1 < \varphi_+,$$

with multiplier $(5a - 9)/(a - 1)$. It is stable iff $\frac{5}{6} < \lambda k < 1$, and it alternates between shading and exaggeration. (iv) The cycle is born by a flip at $a = 2$ and loses stability by a second flip at $a = \frac{5}{3}$.

Relation to prior work. Bénabou & Laroque (1992) derive credibility dynamics from Bayesian learning about a random honest type, and Kartik et al. (2007) study costly talk with credulous receivers. Our contribution is a trading-coupled linear-quadratic model in which the false-alarm region, the Say-Do covariance identity and the bifurcation thresholds are all explicit and estimable.

5. Measuring the Voices

5.1. Echo: exact semantic declustering

Articles arrive at times $\tau_1 < \dots < \tau_N$ with embeddings $\mathbf{z}_j$ and a Hawkes intensity $\lambda(t) = \mu + \sum_{\tau_l < t} g(t - \tau_l)$ with $n = \int g < 1$. In the cluster representation (Hawkes & Oakes, 1974), each article is either news (an immigrant) or an echo of an earlier article.

Proposition 5.1 (Echo share). *In stationarity, the long-run share of articles that are echoes equals the branching ratio n , and the mean cascade size is $(1 - n)^{-1}$.*

Assumption 5.2 (Marked cascade). News embeddings are i.i.d. with density f_0 . An echo of article l has density $f(\cdot | \mathbf{z}_l)$, for example von Mises-Fisher, $f \propto e^{\kappa \langle \mathbf{z}, \mathbf{z}_l \rangle}$.

Theorem 5.3 (Semantic declustering). *Given all times and embeddings on $[0, T]$, the parent labels $\pi(j)$ are independent, with*

$$\mathbb{P}(\pi(j) = 0 | \cdot) = \frac{\mu f_0(\mathbf{z}_j)}{\Lambda_j},$$

$$\mathbb{P}(\pi(j) = l | \cdot) = \frac{g(\tau_j - \tau_l) f(\mathbf{z}_j | \mathbf{z}_l)}{\Lambda_j},$$

where $\Lambda_j = \mu f_0(\mathbf{z}_j) + \sum_{l < j} g(\tau_j - \tau_l) f(\mathbf{z}_j | \mathbf{z}_l)$. The MMSE split of sentiment $s_j = \langle \mathbf{w}_s, \mathbf{z}_j \rangle$ is $s_j^{\text{new}} = \sum_j p_j^{\text{sem}} s_j$ and $s_j^{\text{echo}} = \sum_j (1 - p_j^{\text{sem}}) s_j$, with $p_j^{\text{sem}} = \mu f_0(\mathbf{z}_j) / \Lambda_j$. Semantics never increases the expected posterior entropy of parentage relative to timing alone.

Setting $f \equiv f_0$ recovers timing-only stochastic declustering (Zhuang et al., 2002), where $p_j = \mu / \lambda(\tau_j)$. Combined with Corollary 3.4, s^{echo} estimates exactly the garbled component that the crowd should not have priced (Figure 5).

5.2. Market-aligned embeddings

Documents are embedded on the sphere, $\mathbf{z}_j \in S^{D-1}$. Each has a post-event response vector $\boldsymbol{\rho}_j$ (multi-horizon abnormal returns, abnormal volume, change in implied volatility). Let $S_{jk} = \langle \mathbf{z}_j, \mathbf{z}_k \rangle$ and $D_{jk} = \|\boldsymbol{\rho}_j - \boldsymbol{\rho}_k\|^2$. The return-aligned soft-contrastive loss is $\mathcal{L}(Z) = -\sum_j \sum_{k \neq j} q_{jk} \log p_{jk}(Z)$, where

$$q_{jk} \propto e^{-D_{jk}/2b^2}, \quad p_{jk}(Z) \propto e^{S_{jk}/\tau},$$

and both are normalised over $k \neq j$.

Theorem 5.4 (Minimisers are isometries of outcome space). $\mathcal{L}(Z) \geq \sum_j H(q_j)$, with equality iff $S_{jk} = c - \frac{\tau}{2b^2} D_{jk}$ for some constant c and all $j \neq k$. At equality, $\|\mathbf{z}_j - \mathbf{z}_k\|^2 = 2(1 - c) + \frac{\tau}{b^2} \|\boldsymbol{\rho}_j - \boldsymbol{\rho}_k\|^2$, so embedding neighbours are exactly outcome neighbours.

Theorem 5.5 (Robust neighbour preservation). *For any Z , let $\epsilon_j = \text{KL}(q_j \| p_j(\cdot | Z))$. If $q_{jk} - q_{jl} > \sqrt{2\epsilon_j}$, then $\|\mathbf{z}_j - \mathbf{z}_k\| < \|\mathbf{z}_j - \mathbf{z}_l\|$.*

Theorem 5.5 turns the leftover training loss into a certificate of which analog rankings are reliable. This justifies “déjà vu” forecasting of the narrative-implied reaction by averaging outcomes of the nearest historical analogues (Appendix B). Sentiment is read as a projection onto a topic-orthogonal direction $\mathbf{w}_s^\perp = (I - UU^\top) \mathbf{w}_s$, which is invariant to shifts along topic directions U . Surprise is measured against an expected embedding $\hat{\mathbf{z}}$ as a Mahalanobis distance, which is χ_D^2 -calibrated (Figure 6).

5.3. Who moved first? Lévy area

For paths X, Y , the Lévy area $\mathcal{A}_T = \frac{1}{2} \int_0^T (X_u - X_0) dY_u - (Y_u - Y_0) dX_u$ is the antisymmetric part of the level-two signature. For a closed loop it is the signed enclosed area.

Theorem 5.6 (Expected area identifies lead-lag). *Let (X, Y) be zero-mean, jointly stationary and mean-square C^1 , with cross-covariance $C(h) = \mathbb{E}[X_s Y_{s+h}] \in C^1$. Then*

$$\mathbb{E}[\mathcal{A}_T] = T C'(0) - \frac{1}{2} (C(T) - C(-T)),$$

so $\mathbb{E}[\mathcal{A}_T]/T \rightarrow C'(0)$ when C is bounded. If $Y_t = \gamma X_{t-\ell} + Z_t$, with Z independent and X having a unimodal autocovariance, then the limiting rate has the sign of $\gamma\ell$.

With X = positioning and Y = headline gloom, a positive area rate means that money moved first and the story turned against it (Figure 7).

6. From Signals to Decisions

Events belong to latent regimes (informational, overreaction, strategic) with emissions $\mathbf{o}_t = (s^{\text{new}}, s^{\text{echo}}, G, A, \mathcal{A}, \dots)$.

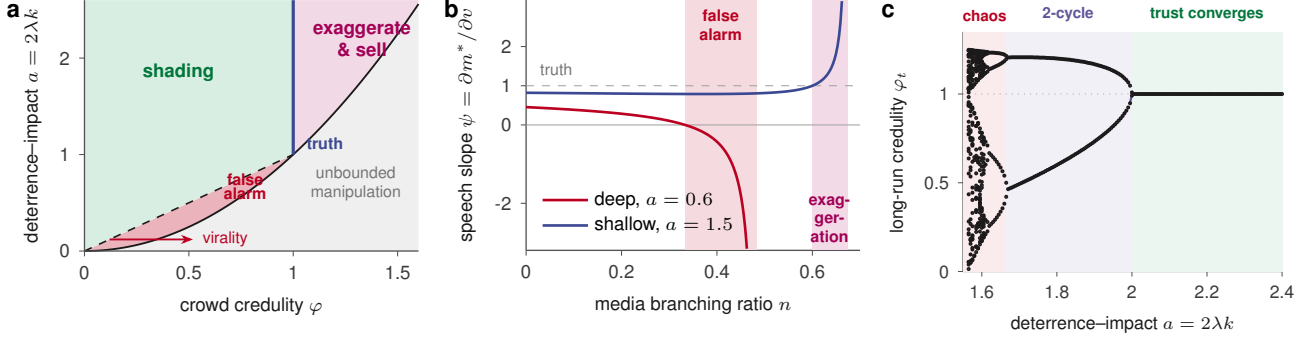


Figure 4. The economics of false alarms. **a**, Regime map (Theorem 4.3 and Corollary 4.4). False alarms occupy the wedge $\varphi^2 < a < \varphi$; echo amplification moves the market to the right. **b**, Speech slope against the media echo share (Proposition 4.6, $\varphi_0 = 0.4$). Deep markets pass through a false-alarm window ($\psi < 0$); shallow markets pass through an exaggeration window ($\psi > 1$). **c**, Bifurcation diagram of adaptive trust (Theorem 4.7). Dots show the long-run attractor, and dashed curves the closed-form cycle $\varphi_{\pm}$. Trust converges for $a > 2$, cycles for $\frac{5}{3} < a < 2$, and becomes chaotic below (numerical Lyapunov exponent ≈ 0.10 at $a = 1.6$).

Table 1. Theory versus simulation. Theory values are exact; simulations use 2×10^6 draws (seed 0); ψ, χ at $\varphi = 0.6, a = 0.5$.

Result	Quantity	Theory	Sim.	Result	Quantity	Theory	Sim.
Corollary 3.4	Bayes weight on echo	0	-0.0006	Theorem 4.7	cycle $\varphi_+, a=1.95$	1.131	1.131
Theorem 3.3	loading, echo channel	-0.1840	-0.1843	Theorem 4.7	multiplier, $a=1.8$	0	superstable
Theorem 3.6	loading on $\hat{x}$ (c_x)	0.4012	0.4004	Theorem 5.3	max posterior error	0	1.1×10^{-16}
Theorem 3.7	absorption coefficient c_A	-0.1997	-0.2004	Theorem 5.6	area rate, lag $\ell = 0.5$	0.2356	0.2360
Theorem 4.3	speech slope ψ	-0.714	-0.714	Theorem B.4	growth at $c^* = 0.338$	0.01056	0.01055
Theorem 4.5	$\text{Cov}(r, \tilde{m}), t_3$ values	-2.294	-2.293	Theorem B.4	growth at full Kelly	-0.0300	-0.0300
Theorem 4.5	same, exaggeration	-7.608	-7.676	Theorem 4.3	trade slope χ	1.429	1.429

Table 2. Illustration: a hypothetical downgrade (numbers are illustrative, not data).

Signal (result)	Reading
news / echo sentiment (Thm 5.3)	$-0.5\sigma / -2.3\sigma$
echo share n (Prop. 5.1)	0.82
Say-Do covariance (Thm 4.5)	negative: fade Say
implied vs actual move (Thm 3.7)	-6.0% vs -1.9%
Lévy area rate (Thm 5.6)	positive: money led
regime I / O / S	$0.10 / 0.28 / 0.62$
Kelly fraction (Thm B.4), $\text{SNR} \approx 0.5$	$c^* \approx 0.33$
Reading: probable false alarm; lean long at one third of full Kelly.	

Filtered regime probabilities are causal, and the MSE-optimal forecast is the mixture of regime experts. Strategic episodes of n events are detected with error at most $\frac{1}{2}e^{-n\Delta^2/8}$. Adaptive conformal intervals attain long-run coverage deterministically (Gibbs & Candès, 2021). Positions use the shrunk Kelly fraction $c^* = \text{SNR}/(1 + \text{SNR})$, since full Kelly destroys growth when the edge is noisier than it is large (Appendix B). A transparent composite whose every sign is fixed by a theorem is

$$D = \omega_1 s^{\text{new}} - \omega_2 s^{\text{echo}} + \omega_3 \text{Do} + \widehat{\text{sgn}} \omega_4 \gamma \text{Say} + \omega_5 A + \omega_6 \mathcal{A}^{\pm}.$$

Here $\widehat{\text{sgn}}$ is the rolling sign of $\text{Cov}(\text{Say}, \text{Do})$ (Theorem 4.5): deeds are always followed, and words are followed when the covariance is positive and faded when it is negative, in which case the two terms combine into $-G_\gamma$ (Theorem 3.6). $\mathcal{A}^{\pm}$ is the Lévy area signed by the direction of positioning. Figure 8 shows the complete architecture.

7. Numerical Verification

Every closed form was checked independently (Table 1). Gaussian results were checked by regression on 2×10^6 simulated draws. Theorem 4.5 was checked with fat-tailed values (t_3) and Laplace noise, which tests its distribution-free claim. The optimal speech of Theorem 4.3 was checked against numerical optimisation. Theorem 5.3 was checked by brute-force enumeration of all parent labellings, and Theorem 5.6 on a smooth Gaussian process with a known lag.

8. Discussion

Testable predictions. The theory predicts:

- opposite signs on s^{new} and s^{echo} ;
- a negative Say-Do covariance in episodes later identified as manipulative;
- false alarms concentrated at intermediate echo shares in

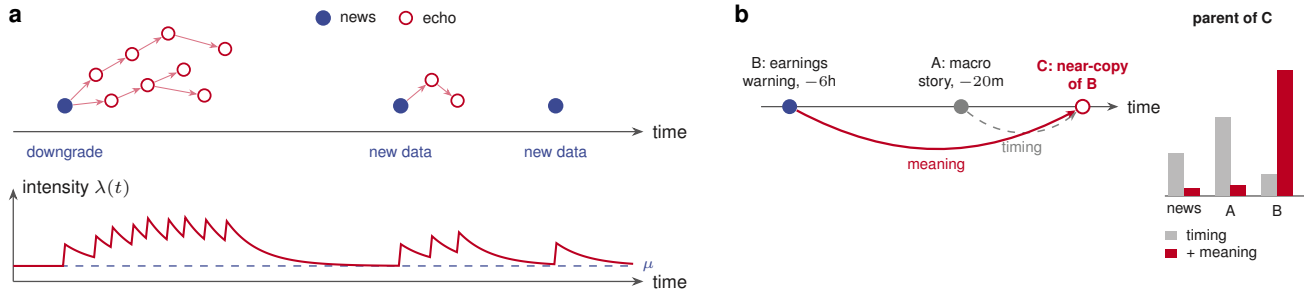


Figure 5. **Separating news from echo.** **a**, A news article (blue) triggers a cascade of echoes (red), and the intensity jumps and decays to the baseline μ . The long-run echo share equals the branching ratio (Proposition 5.1). **b**, Semantic declustering (Theorem 5.3; illustrative numbers). Timing alone attributes article C to the most recent story A; adding embedding similarity correctly identifies C as an echo of B.

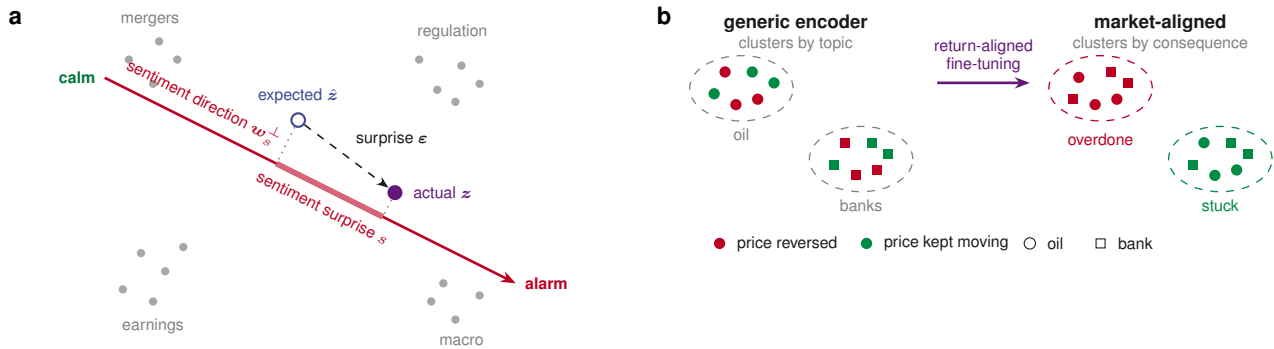


Figure 6. **A map of meaning aligned with markets.** **a**, Embeddings cluster by topic, sentiment is a direction, and surprise is measured against the *expected* story; its projection onto the alarm direction is the sentiment surprise. **b**, At the optimum of the return-aligned objective (Theorem 5.4), distances mirror market aftermaths, so “news that got overdone” becomes a region of the map.

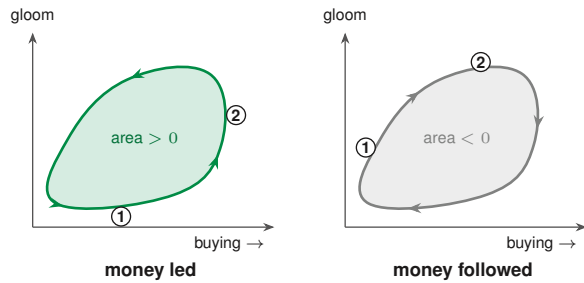


Figure 7. **Lead-lag as a loop.** Traced in the (positioning, gloom) plane, the orientation of the path shows who moved first, and its enclosed area shows how strongly (Theorem 5.6).

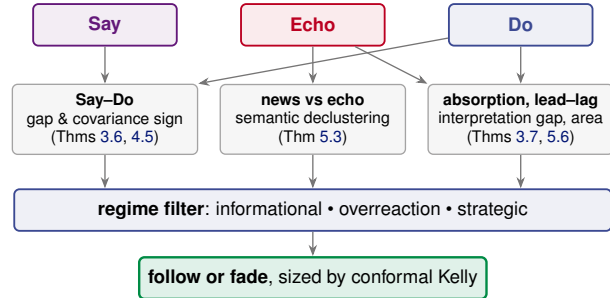


Figure 8. **Architecture.** Three voices produce three families of theorem-signed signals, and a regime filter turns them into a single calibrated decision.

liquid markets, and exaggeration in illiquid ones;

- negative autocorrelation of the rolling Say–Do covariance where λk is moderate;
- an absorption loading whose sign flips with flow toxicity;
- higher returns after positive Lévy area.

The cleanest first test pairs management tone on earnings calls (**Say**) with insider trades (**Do**), both of which are public.

Limitations. Crowd pricing is reduced-form, and credulity is exogenous within a round. The price-formation results fix signs and sufficient statistics, not magnitudes. Theorem 5.6 assumes smooth, stationary paths. Empirical validation with point-in-time data and purged cross-validation (López de Prado, 2018; Bailey & López de Prado, 2014) is left to companion work.

Impact Statement

This paper develops theory for reading the disagreement between institutional statements, media coverage and capital allocation in financial markets. A “strategic” reading means only that narrative and capital moved in opposite directions in a particular order; it is not evidence of deception by any party, and firms maintain information barriers between research and trading. The framework is designed to use public information and lawful disclosures only. The same analysis that detects manipulative communication could in principle inform it, so we stress that creating or spreading narratives intended to move prices is market manipulation, and the methods should be used for detection, surveillance and investor protection. Large language models were used to assist in drafting this manuscript; all content was reviewed by the authors, who take full responsibility for it.

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A. Proofs

Lemma A.1 (Gaussian conditioning). *If $(\mathbf{a}, \mathbf{b})$ is jointly Gaussian with $\text{Var } \mathbf{b} \succ 0$, then $\mathbb{E}[\mathbf{a} \mid \mathbf{b}] = \mathbb{E}\mathbf{a} + \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1}(\mathbf{b} - \mathbb{E}\mathbf{b})$ and $\text{Var}(\mathbf{a} \mid \mathbf{b}) = \text{Var } \mathbf{a} - \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1} \text{Cov}(\mathbf{b}, \mathbf{a})$. The same holds conditionally on a further jointly Gaussian vector.*

Proof. The residual $\mathbf{e} = \mathbf{a} - \mathbb{E}\mathbf{a} - \text{Cov}(\mathbf{a}, \mathbf{b}) \text{Var}(\mathbf{b})^{-1}(\mathbf{b} - \mathbb{E}\mathbf{b})$ is uncorrelated with $\mathbf{b}$, and therefore independent of it by joint Gaussianity. Its variance expands to the stated formula. $\square$

Price and return. From $x = \beta(v - p)$, $p(1 + \lambda\beta) = \boldsymbol{\varphi}^\top \mathbf{y} + \lambda\beta v + \lambda u$, so $r = v - p = \kappa(v - \boldsymbol{\varphi}^\top \mathbf{y} - \lambda u)$.

Proof of Theorem 3.3. $\mathbb{E}[v \mid \mathbf{y}] = \sigma_v^2 \mathbf{h}^\top \Sigma_y^{-1} \mathbf{y} = \mathbf{w}^{\star\top} \mathbf{y}$ and $\mathbb{E}[u \mid \mathbf{y}] = 0$. The residual $r - \mathbb{E}[r \mid \mathbf{y}] = \kappa((v - \mathbb{E}[v \mid \mathbf{y}]) - \lambda u)$ has independent components given $\mathbf{y}$. $\square$

Proof of Corollary 3.4. $\sigma(\mathbf{y}) = \sigma(\mathbf{y}_{-e}, \xi)$ with ξ independent of $(v, \mathbf{y}_{-e})$, so $\mathbb{E}[v \mid \mathbf{y}] = \mathbb{E}[v \mid \mathbf{y}_{-e}]$, which is linear in $\mathbf{y}_{-e}$ alone. If $\mathbf{w}^{\star\top} \mathbf{y} = \mathbf{w}'^\top \mathbf{y}$ almost surely with $w'_e = 0$, then $(\mathbf{w}^* - \mathbf{w}')^\top \Sigma_y (\mathbf{w}^* - \mathbf{w}') = 0$, and $\Sigma_y \succ 0$ gives $\mathbf{w}^* = \mathbf{w}'$. $\square$

Proof of Corollary 3.5. Conditionally on $\mathbf{y}_{-n}$, $v \sim \mathcal{N}(\mu', s^2)$ with $s^2 > 0$. The scalar update puts weight $s^2/(s^2 + \sigma_n^2) \in (0, 1)$ on n . Apply Theorem 3.3. $\square$

Proof of Theorem 3.6. $x = \beta r$, so $\hat{x} = \beta r + \xi_x$. Given $\mathbf{y}$, $\text{Cov}(r, \hat{x}) = \beta \zeta^2$ and $\text{Var } \hat{x} = \beta^2 \zeta^2 + \sigma_x^2$. Lemma A.1 gives $\mathbb{E}[r \mid \mathbf{y}, \hat{x}] = (1 - \beta c_x) \mathbb{E}[r \mid \mathbf{y}] + c_x \hat{x}$ with $1 - \beta c_x = \theta$, and $\text{Var} = \zeta^2 - c_x \beta \zeta^2 = \theta \zeta^2$. The $(m, \hat{x})$ terms are $-\theta \kappa(\varphi_1 - w_1^*)m + c_x \hat{x} = -c_x(\gamma m - \hat{x})$. $\square$

Proof of Theorem 3.7. Let $\tilde{v} = v - \mathbb{E}[v \mid \mathbf{y}]$. Then $r - \mathbb{E}[r \mid \mathbf{y}] = \kappa(\tilde{v} - \lambda u)$. Since $a = \lambda \beta r + \lambda u$, we get $a - \mathbb{E}[a \mid \mathbf{y}] = \lambda \beta \kappa \tilde{v} + \lambda(1 - \lambda \beta \kappa)u = \lambda \kappa(\beta \tilde{v} + u)$, using $1 - \lambda \beta \kappa = \kappa$. Hence $\text{Cov}(r, a \mid \mathbf{y}) = \lambda \kappa^2(\beta \sigma_{v|\mathbf{y}}^2 - \lambda \sigma_u^2)$ and $\text{Var}(a \mid \mathbf{y}) = \lambda^2 \kappa^2(\beta^2 \sigma_{v|\mathbf{y}}^2 + \sigma_u^2)$. Their ratio is c_A by Lemma A.1. At Kyle's intensities, $\beta \sigma_{v|\mathbf{y}}^2 = \sigma_u \sigma_{v|\mathbf{y}}$ and $\lambda \sigma_u^2 = \frac{1}{2} \sigma_u \sigma_{v|\mathbf{y}}$, which give $c_A = \frac{1}{2}$. $\square$

Proof of Lemma 3.8. $\hat{a} = a + (\boldsymbol{\varphi} - \hat{\boldsymbol{\varphi}})^\top \mathbf{y}$ is an invertible affine map of a given $\mathbf{y}$. $\square$

Proof of Theorem 4.3. J is quadratic with Hessian $H = \begin{pmatrix} -2\lambda & -\varphi \\ -\varphi & -k \end{pmatrix}$, which is negative definite iff $\det H = a - \varphi^2 > 0$. Then J is strictly concave, and its unique maximiser solves $v - \varphi m - 2\lambda x = 0$ and $-\varphi x - k(m - v) = 0$. Substitution gives $m(a - \varphi^2) = v(a - \varphi)$, and $1 - \varphi \psi = a(1 - \varphi)/(a - \varphi^2)$ gives χ . If $\det H < 0$, then $J(t\mathbf{d}) \rightarrow \infty$ along an eigenvector $\mathbf{d}$ with positive eigenvalue. $\square$

Proof of Corollary 4.4. $\psi > 0 \iff a > \varphi$; $\psi < 1 \iff \varphi < 1$; and $\text{sgn } \chi = \text{sgn}(1 - \varphi)$. If $\varphi > 1$, then $a > \varphi^2 > \varphi$, so $\psi > 1$. $\square$

Proof of Theorem 4.5. Given m , the optimal trade is $x = (v - \varphi m)/(2\lambda)$, so $v - \varphi m = 2\lambda x$ and $r = v - \varphi(m + \varepsilon) - \lambda x = \lambda x - \varphi \varepsilon$. Since x and m are functions of v , which is independent of ε , we have $\text{Cov}(\varepsilon, \tilde{m}) = \sigma_\varepsilon^2$ and $\mathbb{E}[\varepsilon \mid x] = 0$. In equilibrium, $\text{Cov}(\chi v, \psi v) = \psi \chi \sigma_v^2$, and $\psi \chi < 0$ exactly in the false-alarm and exaggeration regimes. $\square$

Proof of Proposition 4.6. $\varphi = \varphi_0/(1 - n)$ increases from φ_0 to ∞ . A false alarm needs $\varphi^2 < a < \varphi < 1$, hence $a < 1$. Exaggeration needs $1 < \varphi < \sqrt{a}$, hence $a > 1$. The conditions $\varphi > a$, $\varphi > 1$ and $\varphi < \sqrt{a}$ translate to $n > 1 - \varphi_0/a$, $n > 1 - \varphi_0$ and $n < 1 - \varphi_0/\sqrt{a}$. For $a < 1$, $\varphi < \sqrt{a}$ already implies $\varphi < 1$. $\square$

Proof of Theorem 4.7. (i) $F_a(\varphi) = \varphi \iff \varphi = 1$, and admissibility ($\varphi^2 < a$) needs $a > 1$. (ii) $F'_a(\varphi) = (a - 2a\varphi + \varphi^2)/(a - \varphi)^2$, so $F'_a(1) = -1/(a - 1)$, whose modulus is below 1 iff $a > 2$. (iii) Using $a - F_a(\varphi) = (a^2 - a\varphi - a + \varphi^2)/(a - \varphi)$, direct computation gives

$$F_a(F_a(\varphi)) - \varphi = \frac{-a(\varphi - 1)(2\varphi^2 - 2a\varphi + a^2 - a)}{(a - \varphi)^2(a - F_a(\varphi))}.$$

The quadratic factor has roots $\varphi_{\pm}$, which are real and different from 1 for $1 < a < 2$; its value at $\varphi = 1$ is $(a-1)(a-2) \neq 0$. They are admissible: $\varphi_{\pm}^2 = \frac{a}{2}(1 \pm \sqrt{a(2-a)}) < a$ because $a(2-a) < 1$; $\varphi_{\pm} < a$ because $\sqrt{a(2-a)} < a \iff a > 1$; and $\psi(\varphi_{\pm}) > 0$. On the cycle, $\varphi^2 = a\varphi - (a^2 - a)/2$, so the numerator of F'_a equals $a(3 - a - 2\varphi)/2$. With $\varphi_+ + \varphi_- = a$ and $\varphi_+\varphi_- = (a^2 - a)/2$, we get $(3 - a - 2\varphi_+)(3 - a - 2\varphi_-) = (5a - 9)(a - 1)$ and $(a - \varphi_+)(a - \varphi_-) = a(a - 1)/2$. Hence the multiplier is $(5a - 9)/(a - 1)$, whose modulus is below 1 iff $\frac{5}{3} < a < 2$ when $1 < a < 2$. (iv) At $a = 2$ the multiplier is 1 and $\varphi_{\pm} = 1$; at $a = \frac{5}{3}$ it is -1 . $\square$

Proof of Proposition 5.1. Each news article founds a Galton–Watson cluster of expected size $\sum_k n^k = (1 - n)^{-1}$. News arrives at rate μ , so articles arrive at rate $\mu/(1 - n)$, and the news share is $1 - n$. $\square$

Proof of Theorem 5.3. In the cluster representation, a labelled configuration is a realisation of independent Poisson streams: news at rate $\rho_0 = \mu$, and echoes of l at rate $\rho_l(t) = g(t - \tau_l)\mathbf{1}\{t > \tau_l\}$. Its density is the product of the stream likelihoods and the mark densities,

$$e^{-\int_0^T \lambda} \prod_j \rho_{\pi(j)}(\tau_j) f_{\pi(j)}(\mathbf{z}_j), \quad f_0(\text{news}), \quad f_l = f(\cdot \mid \mathbf{z}_l),$$

because $\sum_k \rho_k = \lambda$. This factorises over j . Summing over labellings gives $e^{-\int \lambda} \prod_j \Lambda_j$, so the posterior is a product of the stated categorical laws. Because s_j is a function of $\mathbf{z}_j$, $\mathbb{E}[\mathbf{1}\{\pi(j) = 0\} s_j \mid \cdot] = s_j p_j^{\text{sem}}$. Integrating out the marks recovers the timing-only posterior, and conditioning does not increase expected entropy. $\square$

Proof of Theorem 5.4. Per anchor, $-\sum_k q_{jk} \log p_{jk} = H(q_{j\cdot}) + \text{KL}(q_{j\cdot} \parallel p_{j\cdot})$, with equality to $H(q_{j\cdot})$ iff $p_{j\cdot} = q_{j\cdot}$, that is, iff the logits differ by a constant c_j : $S_{jk}/\tau + D_{jk}/2b^2 = c_j/\tau$. Symmetry of S and D forces $c_j = c_k$ for every pair, and on the sphere $\|\mathbf{z}_j - \mathbf{z}_k\|^2 = 2 - 2S_{jk}$. $\square$

Proof of Theorem 5.5. Pinsker gives $\sum_k |p_{jk} - q_{jk}| \leq \sqrt{2\epsilon_j}$, so $p_{jk} - p_{jl} \geq q_{jk} - q_{jl} - \sqrt{2\epsilon_j} > 0$. Softmax is monotone in $S_{j\cdot}$, so $S_{jk} > S_{jl}$, which gives the distance ordering. $\square$

Proof of Theorem 5.6. Mean-square differentiability gives $\mathbb{E}[X_s \dot{Y}_t] = C'(t - s)$ and $\mathbb{E}[Y_s \dot{X}_t] = -C'(s - t)$. Hence $\mathbb{E}[(X_u - X_0)\dot{Y}_u] = C'(0) - C'(u)$ and $\mathbb{E}[(Y_u - Y_0)\dot{X}_u] = -C'(0) + C'(-u)$. By Fubini,

$$\mathbb{E}[\mathcal{A}_T] = \frac{1}{2} \int_0^T (2C'(0) - C'(u) - C'(-u)) du = TC'(0) - \frac{1}{2}(C(T) - C(-T)).$$

For $Y_t = \gamma X_{t-\ell} + Z_t$, $C(h) = \gamma R(h - \ell)$, so $C'(0) = -\gamma R'(\ell)$. This has the sign of $\gamma\ell$ when R is even with $R' < 0$ on $(0, \infty)$. $\square$

B. Additional Results

Proposition B.1 (Causal filtering and mixture optimality). *Filtered regime probabilities $\pi_t \propto f(\mathbf{o}_t) \odot P^\top \pi_{t-1}$ are $\mathcal{F}_t$ -measurable, and $\mathbb{E}[r_{t+h} \mid \mathcal{F}_t] = \sum_k \pi_t(k) \mathbb{E}[r_{t+h} \mid \mathcal{F}_t, K_t = k]$. Smoothed probabilities are not $\mathcal{F}_t$ -measurable in general and must not be used for forecasting.*

Proof. The recursion is Bayes' rule, and the mixture follows from the tower property. $\square$

Theorem B.2 (Exponential detectability). *For n events that are i.i.d. within a regime, with emissions $\mathcal{N}(\mathbf{m}_i, \Sigma)$ and priors π_i , the MAP error satisfies $P_e \leq \sqrt{\pi_0 \pi_1} e^{-n\Delta^2/8} \leq \frac{1}{2} e^{-n\Delta^2/8}$, where $\Delta^2 = (\mathbf{m}_1 - \mathbf{m}_0)^\top \Sigma^{-1} (\mathbf{m}_1 - \mathbf{m}_0)$.*

Proof. $\min\{a, b\} \leq \sqrt{ab}$ gives $P_e \leq \sqrt{\pi_0 \pi_1} \text{BC}^n$ (Bhattacharyya, 1943; Kailath, 1967). For equal-covariance Gaussians, completing the square gives $\text{BC} = \int \sqrt{p_0 p_1} = e^{-\Delta^2/8}$. $\square$

Theorem B.3 (Adaptive conformal coverage; Gibbs & Candès, 2021, Prop. 4.1). *With nested sets $\hat{C}_t(\alpha')$ that equal $\mathbb{R}$ for $\alpha' \leq 0$ and $\emptyset$ for $\alpha' \geq 1$, and the update $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, every outcome sequence satisfies $|\frac{1}{T} \sum_t \text{err}_t - \alpha| \leq (\max\{\alpha_1, 1 - \alpha_1\} + \gamma)/(\gamma T)$.*

Proof (reproduced). By induction, $\alpha_t \in [-\gamma, 1 + \gamma]$. Telescoping gives $\alpha_{T+1} - \alpha_1 = \gamma \sum_t (\alpha - \text{err}_t)$. $\square$

Theorem B.4 (Kelly shrinkage under estimation risk; classical). *With excess growth rate $g(f) = f\mu - \frac{1}{2}f^2\sigma^2$ (Merton, 1969; Kelly, 1956), an unbiased estimate $\hat{\mu}$ with variance s^2 that is independent of future returns, and $f = c\hat{\mu}/\sigma^2$, the expected growth is $\mathbb{E}g = (c\mu^2 - \frac{1}{2}c^2(\mu^2 + s^2))/\sigma^2$. It is maximised at $c^* = \mu^2/(\mu^2 + s^2)$. Full Kelly yields $(\mu^2 - s^2)/(2\sigma^2)$, which is negative when $s > |\mu|$.*

Proof. Expand $\mathbb{E}[f\mu]$ and $\mathbb{E}[f^2]$ and maximise the resulting concave quadratic in c . $\square$

Proposition B.5 (Analog forecasting). *If $\text{AR}_j = \mu(z_j) + e_j$ with uncorrelated mean-zero errors of variance at most σ^2 , μ is L -Lipschitz, and weights $\omega_j \geq 0$ sum to one, then the estimator $\hat{\mu}(z) = \sum_j \omega_j \text{AR}_j$ has bias at most $L \sum_j \omega_j d(z, z_j)$ and variance at most $\sigma^2 \sum_j \omega_j^2$.*

Proposition B.6 (Calibrated surprise). *If $z \mid \mathcal{F} \sim \mathcal{N}(\hat{z}, \Sigma)$, then $S = (z - \hat{z})^\top \Sigma^{-1} (z - \hat{z}) \sim \chi_D^2$, and S equals Shannon surprisal up to additive constants.*

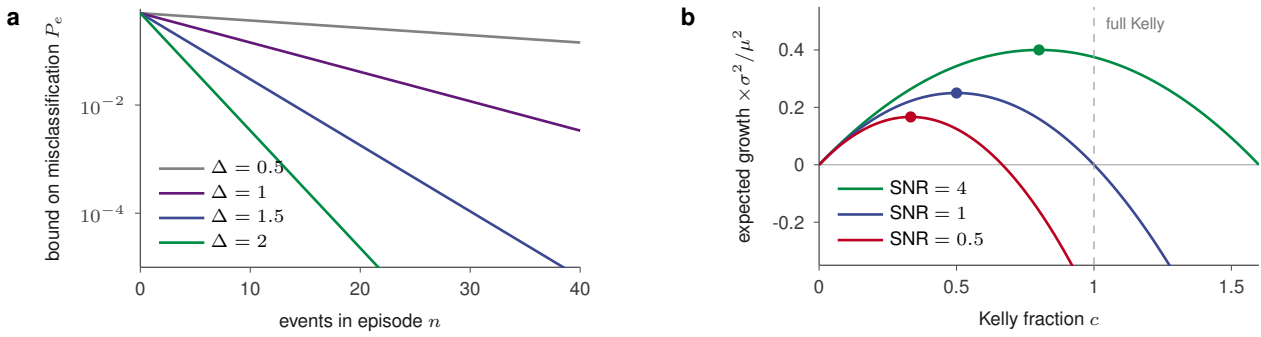


Figure 9. Detection and sizing. **a**, Misclassification of strategic episodes decays exponentially in episode length (Theorem B.2). **b**, Expected growth as a function of the Kelly fraction; dots mark $c^* = \text{SNR}/(1 + \text{SNR})$ (Theorem B.4).

C. Implementation Notes

Encoder. A transformer sentence encoder (Reimers & Gurevych, 2019) with low-rank adapters, trained on the return-aligned loss plus a KL anchor to the frozen encoder, on rolling windows whose outcome horizons close before each refit date. Pretrained components must predate each test window (Glasserman & Lin, 2023).

Echo. Exponential Hawkes kernels are fitted per semantic cluster, with von Mises–Fisher echo marks. The posteriors of Theorem 5.3 cost $O(N \cdot \bar{k})$ with $\bar{k}$ candidate parents retrieved from a vector index.

Say–Do. Pair management tone with insider trades, and published house views with disclosed holdings or futures positioning. A lag-augmented Kalman filter merges disclosures of mixed frequency, and a rolling window estimates $\text{Cov}(\text{Say}, \text{Do})$.

Validation. Point-in-time data, combinatorial purged cross-validation and deflated performance statistics (López de Prado, 2018; Bailey & López de Prado, 2014).